

Options II

The Greeks

Explained so simply a 10-year-old gets it

The five dials — Delta, Gamma, Theta, Vega, Rho — that reveal exactly why an option moves, and the difference between trading options and gambling on them.

Stock Market Mastery Roadmap • Advanced Level
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An option's price wobbles for **several reasons at once** — not just direction. The **Greeks** are five little dials that tell you *exactly* how your option reacts to each force. Learn them and you stop guessing why a trade is bleeding — you *know*. This is the chapter that turns an option **gambler** into an option **trader**.

Delta — your speed with the market

Delta answers: "how much do I make/lose for a 1-point move in the stock?" A delta of 0.5 means you gain about ₹0.50 for every ₹1 the stock rises.

- It's also a **rough probability** of finishing **in-the-money** — a 0.30 delta $\approx$ ~30% chance.
- **Position delta** = the *total* directional exposure of all your legs added up — your true "lean."

Gamma — how fast your speed changes

DELTA IS SPEED, GAMMA IS ACCELERATION

If **delta** is how fast you're moving, **gamma** is how fast that speed itself changes — the **accelerator**.

Gamma is **highest near expiry and at-the-money (ATM)**, where a tiny move can flip your delta wildly.

That's why the last day before expiry feels like driving a race car on ice — thrilling for buyers, terrifying for sellers.

Theta — the daily melt

Theta is **time decay** — how much value your option loses **each day**, just from time passing (the melting ice cream from Chapter 14). It **speeds up** as expiry nears.

- **Buyers pay theta** — they lose a little every day, hoping a big move pays off before time runs out.
- **Sellers collect theta** — they earn that daily melt as income.

Vega — sensitivity to fear & excitement

Vega answers: "how much does my option move for a 1% change in implied volatility (IV)?" IV is the market's **expected drama** — more expected movement = pricier options.

- Vega matters **most in longer-dated options** and **around big events** (earnings, policy), when IV swings a lot.
- Buy options when IV is **low**, be careful buying when IV is **high** and about to fall.

IV CRUSH: THE SNEAKY TRAP

Before a big event, everyone expects fireworks, so **IV rises and options get expensive**. Right *after* the event, the uncertainty vanishes and **IV collapses** — called **IV crush**. You can be **right on direction and still LOSE**, because the vega drop ate more than your delta gained. Never overpay for options into an event without knowing this.

Rho — the quiet one

Rho is sensitivity to **interest rates**. For the short-dated options most Indian traders use, it's **minor** — good to know it exists, rarely worth losing sleep over.

Every position is a bundle of Greeks

Here's the mindset shift: you're not just "buying a call" — you're taking on a **package of Delta + Gamma + Theta + Vega** all at once. Pros **trade the Greeks**, and often **hedge** one away:

- **Delta-hedged / Greek-neutral book** — balancing legs so the position barely cares about small directional moves, isolating the Greek you actually want to bet on (like volatility).
- **Short-gamma** (option sellers) → **profit slowly, lose fast** — they earn theta daily but a sharp move hurts a lot at once.
- **Long-gamma** (option buyers) → the reverse — **bleed slowly (theta), win fast** on a big move.

Volatility: implied vs historical

- **Historical volatility** — how much the stock **actually moved** in the past.

- **Implied volatility (IV)** — how much the market **expects** it to move, baked into option prices.
- **IV term structure** — how IV differs across **expiries** (near vs far).
- **Volatility smile / skew** — different strikes carry different IVs — often puts cost more (crash fear), drawing a "smile" or lopsided "skew" across strikes.

The Greeks at a glance

Greek	Answers the question	Who it favours
Delta	How much do I make per 1-point move?	directional traders
Gamma	How fast does my delta change?	long options (helps buyers, hurts sellers)
Theta	How much do I lose per day to time?	sellers (they collect it)
Vega	How much do I move per 1% IV change?	long options in rising IV
Rho	How much do rates affect me?	minor for short-dated

Key ideas to remember

- **Every option position is a bundle of Greeks** — trade the **Greeks**, not just the direction.
- **Short-gamma profits slowly and loses fast; long-gamma is the reverse.** Sellers sip theta but fear the gap; buyers bleed theta but love the big move.
- **IV crush is real** — after an event, collapsing IV can make **long options lose even when direction was right.**

PRACTICAL EXERCISES

-  For **one ATM option**, record **delta / gamma / theta / vega daily for a week** — watch **gamma and theta rise** into expiry.
-  **Build a spreadsheet** that **sums the position Greeks** for a simple 2-leg spread.
-  **Pick an event** (earnings/policy) and watch **IV rise before and crush after** — measure the premium lost purely to **IV crush**.

REAL-WORLD APPLICATION

-  **Greek awareness is what separates an option trader from a gambler.** It tells you *why* a position is bleeding (theta? vega? gamma?) and *how to fix it* — add a hedge, roll to a new strike/expiry, or cut. Without the Greeks you're flying blind; with them you have a full instrument panel.

ONE-LINE SUMMARY

Five dials drive every option: **Delta** (speed with the market), **Gamma** (acceleration — wild near expiry/ATM), **Theta** (daily melt — sellers collect, buyers pay), **Vega** (IV sensitivity — beware **IV crush**), and **Rho** (rates, minor). Every position is a **bundle of Greeks**: sellers are **short-gamma** (slow gains, fast losses), buyers **long-gamma** (slow bleed, fast wins). Trade the Greeks, not just the guess.